

Molecular Modelling Course

November 18. 2025 – 20. 2025 | Dr. Birkan Emrem, Dr. Ferdinand Jamitzky, Dr. Plamen Dobrev, Dr. Prasanth Ganta, Dr. Elmira Birang
Leibniz Supercomputing Centre

Visualizing biomolecular structures



- PyMOL
- VESTA
- VMD

PyMOL



https://www.pymol.org



PyMOL by Schrödinger

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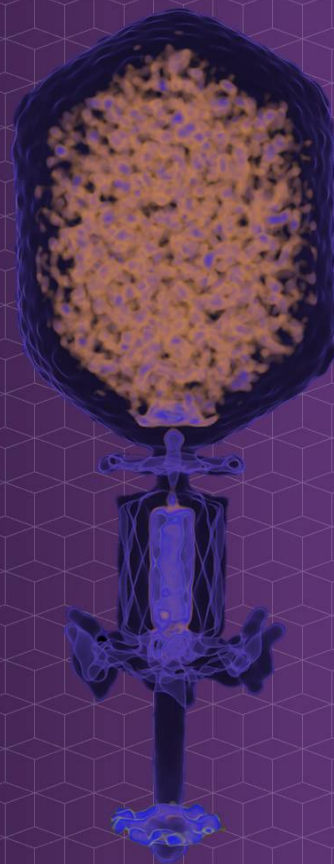
Introducing PyMOL 3.1

PyMOL is a user-sponsored molecular visualization system on an [open-source foundation](#), maintained and distributed by [Schrödinger](#).

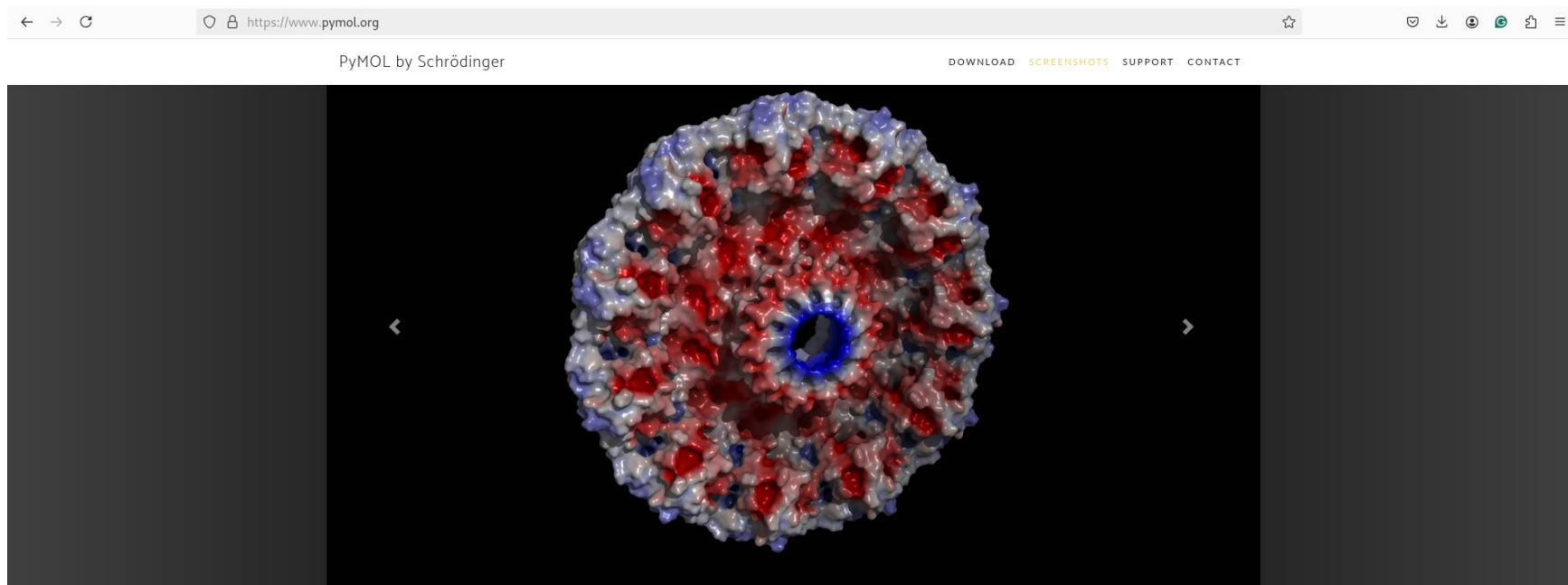
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[Pymol 3 Product Page](#)



PyMOL

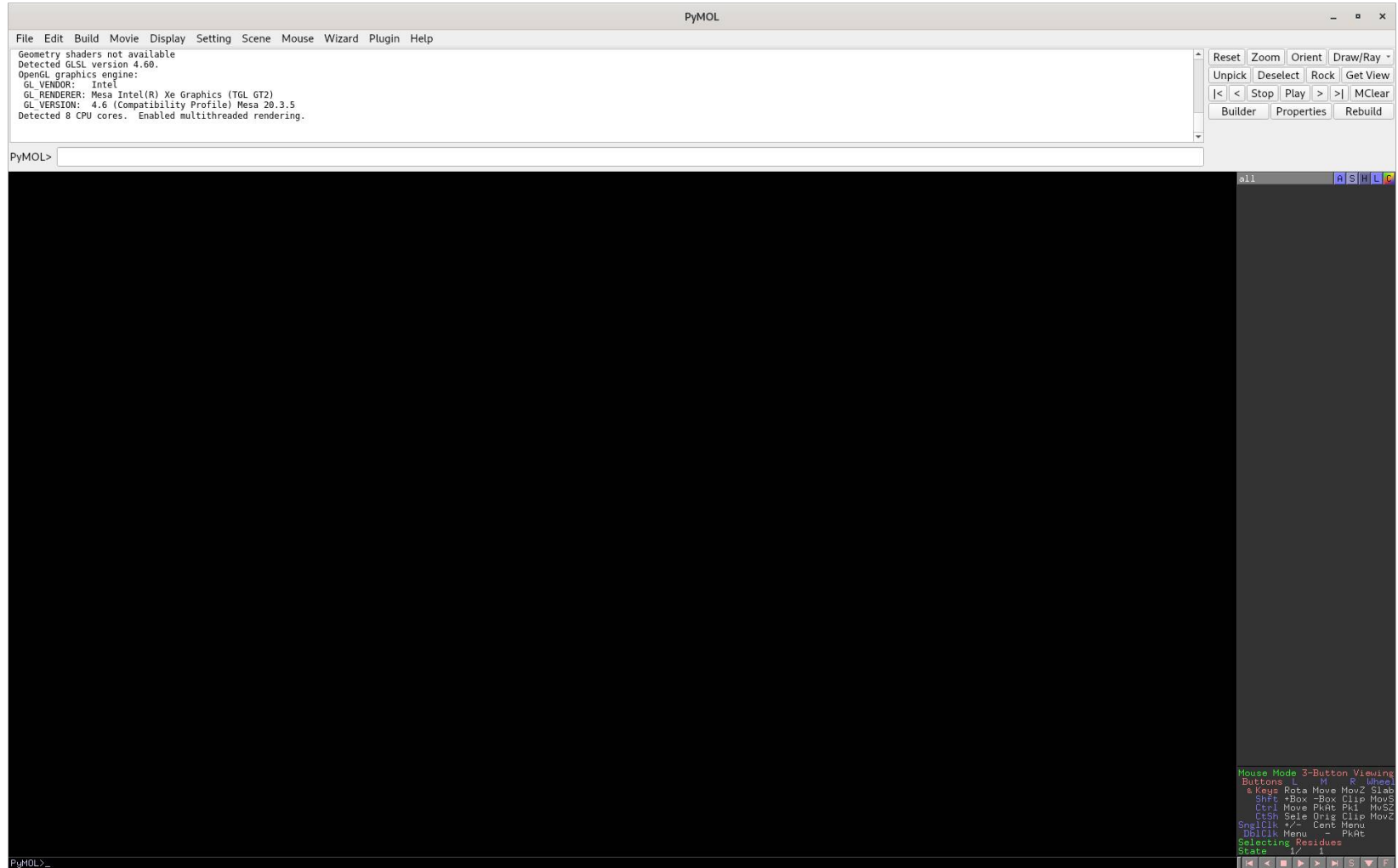


Open-Source Philosophy

PyMOL is a commercial product, but we make most of its source code freely available under a permissive license. The open source project is maintained by **Schrödinger** and ultimately funded by everyone who purchases a PyMOL license.

Open source enables open science.
This was the vision of the original PyMOL author Warren L. DeLano.

PyMOL



- New PyMOL Window
- Open...
- Open Recent...
- Get PDB...
- Save Session
- Save Session As...
- Export Molecule...
- Export Map...
- Export Alignment...
- Export Image As
- Export Movie As
- Log File
- Run Script...
- Working Directory
- Edit pymolrc
- Reinitialize
- Quit

Graphics (TGL GT2)
ity Profile) Mesa 20.3.5
d multithreaded rendering.

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear

Builder Properties Rebuild

all A S H L C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
CtrlSh Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

OpenGL graphics engine:

GL_VENDOR: Intel
GL_RENDERER: Mesa Intel(R) Xe Graphics (TGL GT2)
GL_VERSION: 4.6 (Compatibility Profile) Mesa 20.3.5
Detected 8 CPU cores. Enabled multithreaded rendering.

TITLE LYSOZYME C

ObjectMolecule: Read crystal symmetry information.

CmdLoad: "" loaded as "centered".

Reset Zoom Orient Draw/Ray ▾
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL> 

all
centered 1/2 **MSHLC**

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
s Keys Rota Move MovZ Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MvSZ
CtSh Sele Orig Clip MovZ
SnglClk +/- Cent Menu
DbIClk Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>

OpenGL graphics engine:
GL_VENDOR: Intel
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ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>



all A S H L C

centered Action:
zoom
orient
center
origin
drag matrix
reset matrix
drag coordinates
clean
preset
find
align
generate
assign sec. struc.
rename object
copy to object
group
delete object
hydrogens
remove waters
state
masking
sequence
movement
compute

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

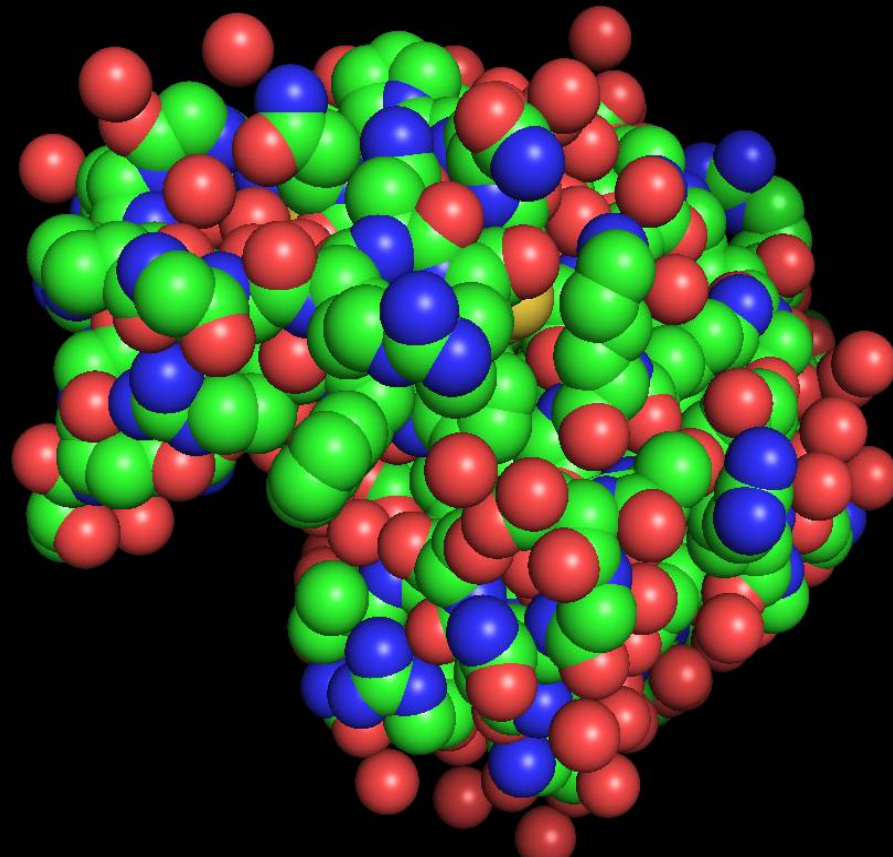
PyMOL>_

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

GL_VENDOR: Intel
GL_RENDERER: Mesa Intel(R) Xe Graphics (TGL GT2)
GL_VERSION: 4.6 (Compatibility Profile) Mesa 20.3.5
Detected 8 CPU cores. Enabled multithreaded rendering.
TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".

Reset Zoom Orient Draw/Ray ▾
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>



all A S H L C
centered 1/1 Show:
as
wire
lines
nonbonded
licorice
sticks
nb_spheres
ribbon
cartoon
label
cell
dots
spheres
mesh
surface
flag ignore
organic
main chain
side chain
disulfides
valence

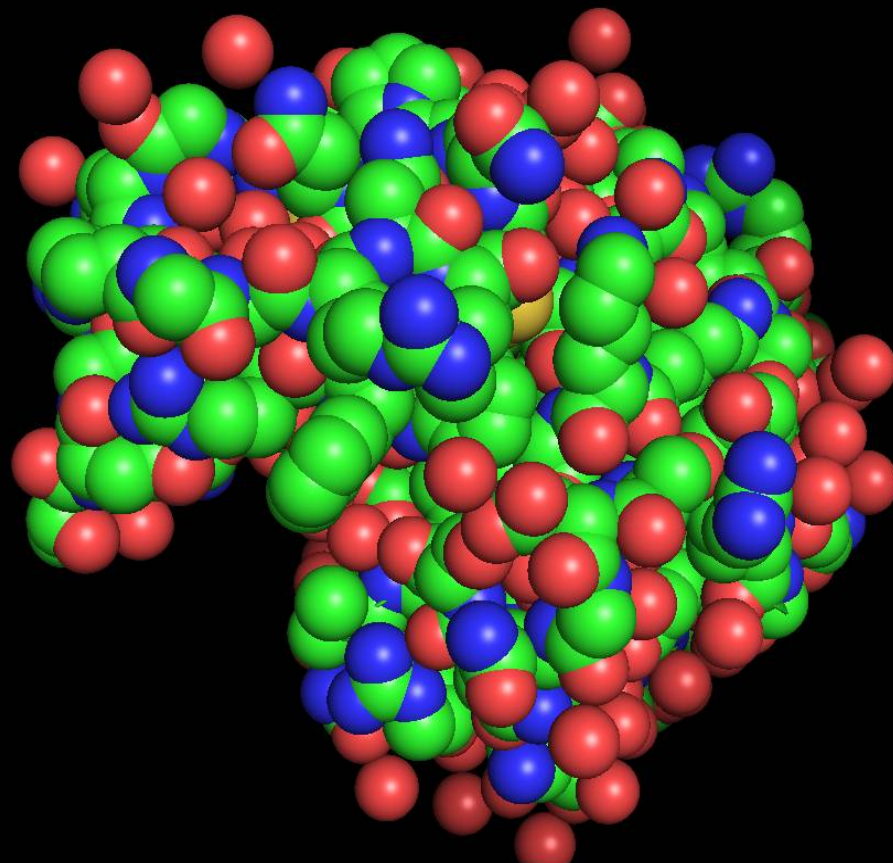
Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

```
COMPND  MOL ID: 1;  
COMPND  2 MOLECULE: LYSOZYME C;  
COMPND  3 CHAIN: A;  
COMPND  4 SYNONYM: 1,4-BETA-N-ACETYLMURAMIDASE C, ALLERGEN GAL D 4, GAL D IV;  
COMPND  5 EC: 3.2.1.17  
ObjectMolecule: Read secondary structure assignments.  
ObjectMolecule: Read crystal symmetry information.  
CmdLoad: "" loaded as "2YVB".
```

PyMOL>

Reset Zoom Orient Draw/Ray ▾
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild



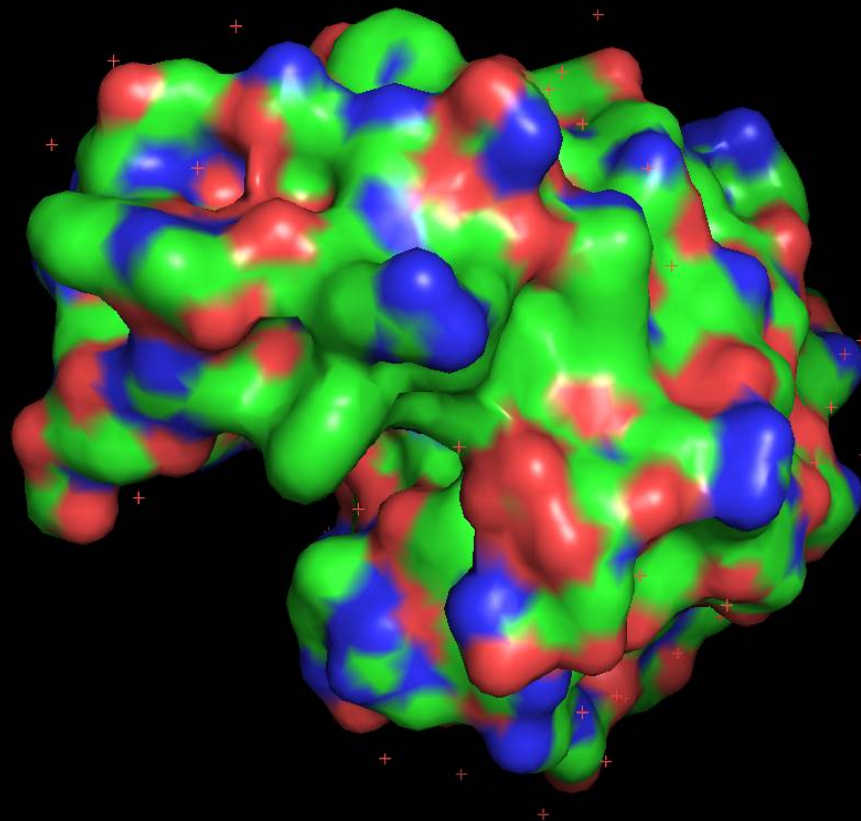
all A S H L C
2YVB 1/1
Hide:
everything
wire
lines
nonbonded
licorice
sticks
nb_spheres
ribbon
cartoon
label
cell
dots
spheres
mesh
surface
flag ignore
main chain
side chain
waters
hydrogens
unselected
valence

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rota Move MovZ Slab
Shift +Box -Box Clip MovS
Ctrl Move PKAt PKI MvSZ
DtSh Sele Drig Clip MovZ
SpicClk +/- Cent Menu
DbiClk Menu - PKAt
Selecting Residues
State 1/ 1

GL_VENDOR: Intel
GL_RENDERER: Mesa Intel(R) Xe Graphics (TGL GT2)
GL_VERSION: 4.6 (Compatibility Profile) Mesa 20.3.5
Detected 8 CPU cores. Enabled multithreaded rendering.
TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>



all A S H L C
centered 1/1 Show:
as
wire
lines
nonbonded
licorice
sticks
nb_spheres
ribbon
cartoon
label
cell
dots
spheres
mesh
surface
flag ignore
organic
main chain
side chain
disulfides
valence

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

GL_VENDOR: Intel
GL_RENDERER: Mesa Intel(R) Xe Graphics
GL_VERSION: 4.6 (Compatibility Profile) Mesa 23.0.3
Detected 8 CPU cores. Enabled multi-threading.
TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".

Edit All...
Keyboard Shortcuts...
Colors...

Label
Lines & Sticks
Cartoon
Ribbon
Surface
Volume

Transparency

Rendering

PDB File Loading

mmCIF File Loading

Map File Loading

Auto-Show ...

☒ Auto-Zoom New Objects

☐ Auto-Remove Hydrogens

☐ Show Text (Esc)

☐ Overlay Text

Surface

Sphere

Cartoon

Stick

Uni-Layer

Multi-Layer

Multi-Layer (Real-time OIT)

Fast and Ugly

☐ Angle-dependent

Off

20%

40%

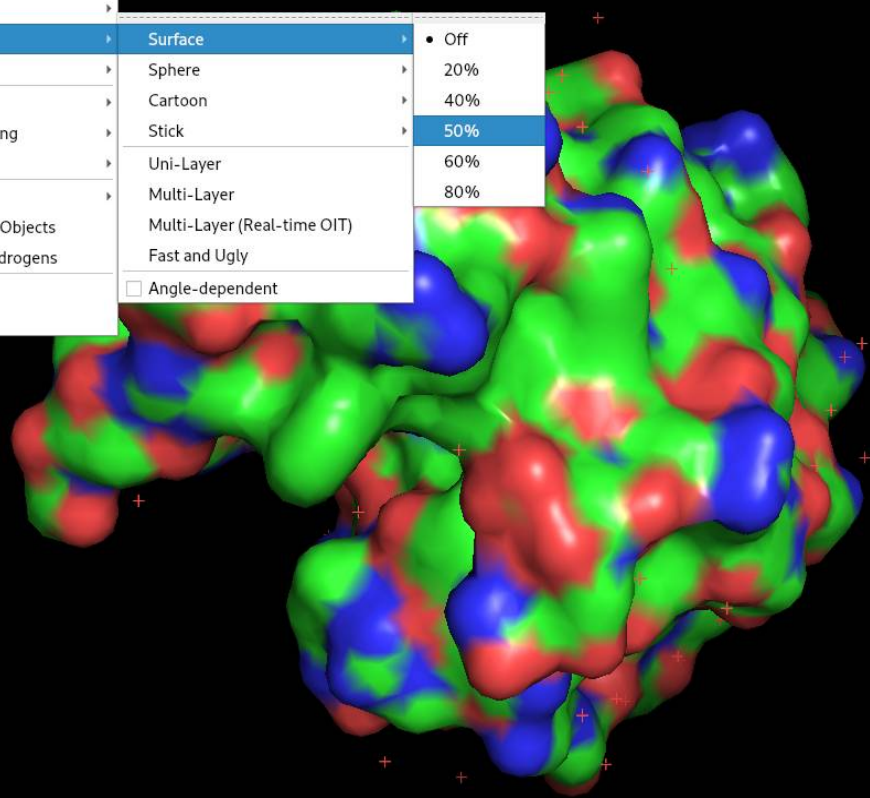
50%

60%

80%

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>



all A S H L C
centered 1/1 A S H L C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

GL RENDERER: Mesa Intel(R) Xe Graphics
GL_VERSION: 4.6 (Compatibility Profile) Mesa 23.0.1
Detected 8 CPU cores. Enabled multi-threading.
TITLE: LYSOZYME C
ObjectMolecule: Read crystal symmetry file
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".
Setting: transparency set to 0.500

PyMOL>

Edit All...
Keyboard Shortcuts...
Colors...

Label
Lines & Sticks
Cartoon
Ribbon
Surface
Volume

Transparency

Rendering

PDB File Loading

mmCIF File Loading

Map File Loading

Auto-Show ...

☒ Auto-Zoom New Objects

☐ Auto-Remove Hydrogens

☐ Show Text (Esc)

☐ Overlay Text

Surface

Sphere

Cartoon

Stick

Uni-Layer

Multi-Layer

Multi-Layer (Real-time OIT)

Fast and Ugly

☐ Angle-dependent

Off

20%

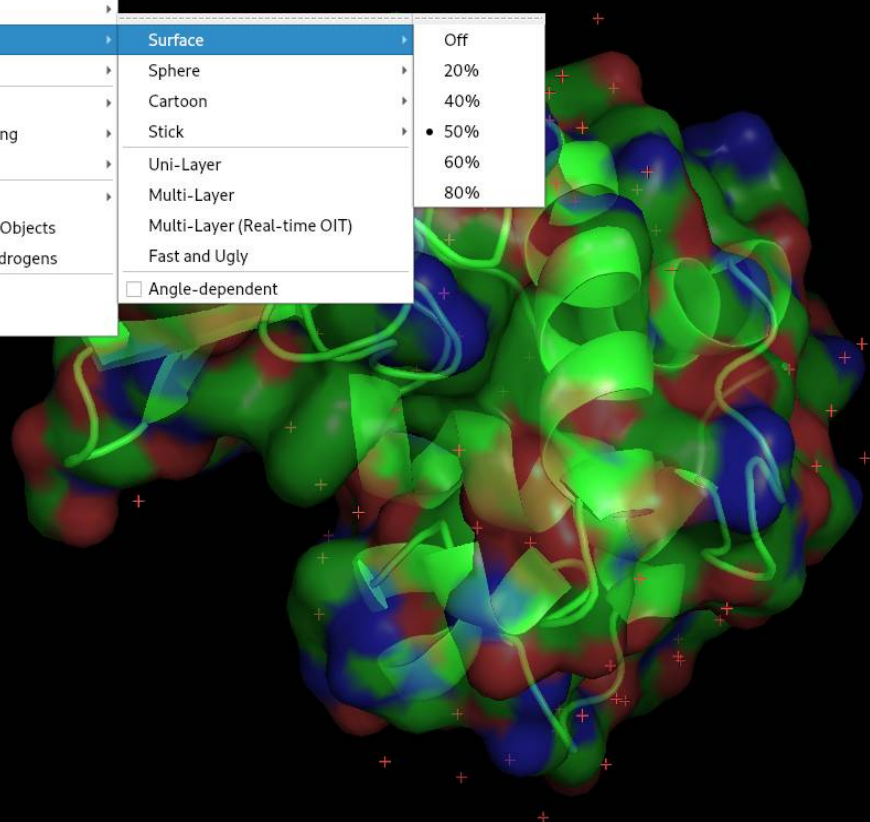
40%

• 50%

60%

80%

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MCLEAR
Builder Properties Rebuild



all A S H L C
centered 1/1 A S H L C

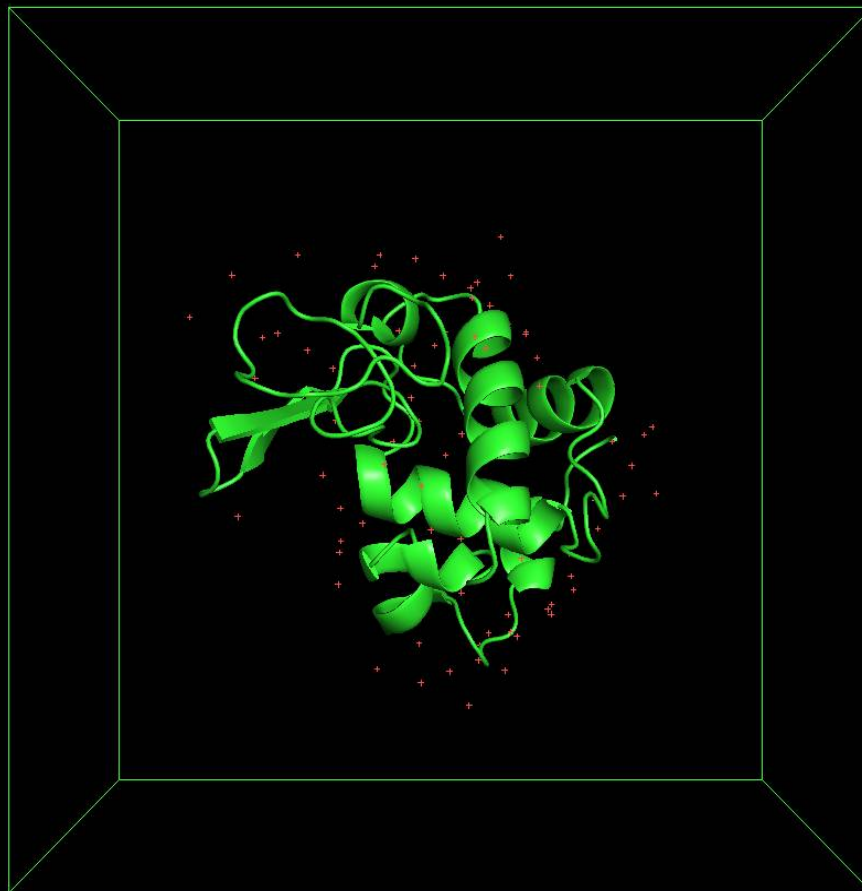
Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rota Move Mov2 Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MvSZ
Ctrl Sele Orig Clip MovZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".
Setting: transparency set to 0.50000.
Spectrum: range (15.29000 to 61.04000).
Executive: Colored 1082 atoms and 1 object.
Spectrum: range (17.45000 to 41.11000).

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>



all A S H L C
centered 1/1 Show:
as
wire
lines
nonbonded
licorice
sticks
nb_spheres
ribbon
cartoon
label
cell
dots
spheres
mesh
surface
flag ignore
organic
main chain
side chain
disulfides
valence

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rota Move Mov2 Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MvSZ
CtSh Sele Orig Clip MovZ
SnglClk +/- Cent Menu
DblClk Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

GL VERSION: 4.6 (Compatibility Profile) Mesa 20.3.5
Detected 8 CPU cores. Enabled multithreaded rendering.
TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".
Setting: transparency set to 0.50000.
Spectrum: range (15.29000 to 61.04000).

Reset	Zoom	Orient	Draw/Ray ▾			
Unpick	Deselect	Rock	Get View			
<	<	Stop	Play	>	>	MClear
Builder		Properties		Rebuild		

PyMOL>



all	A	S	H	L	C
centered 121	Color:				
	by element				
	by chain				
By Secondary Structure:	by ss				
Helix Sheet Loop	by rep				
Helix Sheet Loop	spectrum				
Helix Sheet Loop	auto				
	reds				
	greens				
	blues				
	yellows				
	magentas				
	cyans				
	oranges				
	tints				
	grays				

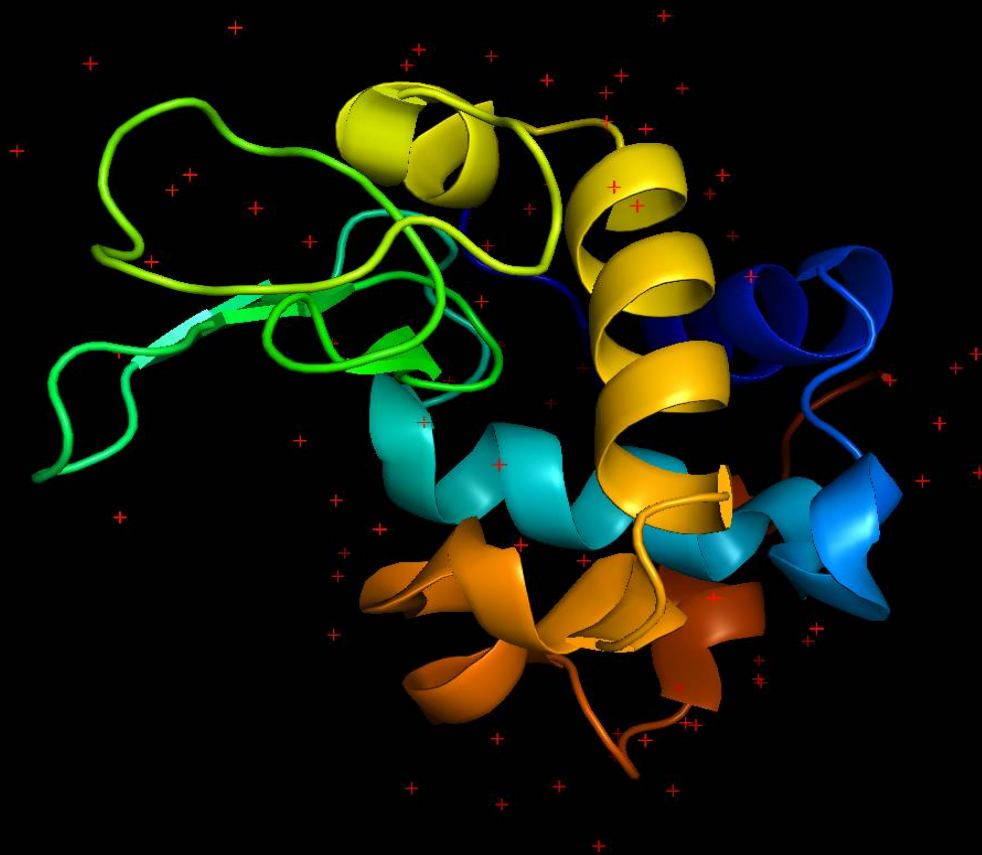
Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

```
COMPND MOL ID: 1;  
COMPND 2 MOLECULE: LYSOZYME C;  
COMPND 3 CHAIN: A;  
COMPND 4 SYNONYM: 1,4-BETA-N-ACETYLMURAMIDASE C, ALLERGEN GAL D 4, GAL D IV;  
COMPND 5 EC: 3.2.1.17  
ObjectMolecule: Read secondary structure assignments.  
ObjectMolecule: Read crystal symmetry information.  
CmdLoad: "" loaded as "2YVB".
```

PyMOL>

Reset Zoom Orient Draw/Ray ▾
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild



all
2YVB 1/1
Color:
by element
by chain
by ss
by rep
Spectrum:
rainbo (elem C) spectrum
rainbo (*CA) auto
rainbow
b-factors
b-factors(*CA)
area (molecular)
area (solvent)
reds
greens
blues
yellows
magentas
cyans
oranges
tints
grays

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
sKeys Rota Move MovZ Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MovSZ
DtSh Sele Orig Clip MovZ
SnapClk +/- Cent Menu
DbClk Menu - PkAt
Selecting Residues
State 1/ 1

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".
Setting: transparency set to 0.50000.
Spectrum: range (15.29000 to 61.04000).
Executive: Colored 1082 atoms and 1 object.
Spectrum: range (17.45000 to 41.11000).

Reset	Zoom	Orient	Draw/Ray ▾
Unpick	Deselect	Rock	Get View
<	<	Stop	Play >
>	MClear		
Builder		Properties	Rebuild

PyMOL>



all	A S H L C
centered 1z1	Color:
	by element
	by chain
	by ss
	by rep
Spectrum:	spectrum
rainbow (elem C)	auto
rainbow (* / CA)	reds
rainbow	greens
b-factors	blues
b-factors (* / CA)	yellows
area (molecular)	magentas
area (solvent)	cyans
	oranges
	tints
	grays

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move Slab
Shift +Box -Box Clip Move
Ctrl Move PkAt Pk1 MvSz
CtSh Sele Orig Clip Move
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

Detected 8 CPU cores. Enabled multithreaded rendering.

TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
Dragging whole object "centered".
Setting: transparency set to 0.50000.
Spectrum: range (15.29000 to 61.04000).
Executive: Colored 1082 atoms and 1 object.

Reset	Zoom	Orient	Draw/Ray ▾
Unpick	Deselect	Rock	Get View
<	<	Stop	Play >
>	>	MClear	
Builder		Properties	Rebuild

PyMOL>



all	A S H L C
centered 1/1	Color:
	by element
	by chain
	by ss
	by rep
	spectrum
Reds	auto
red	reds
tv_red	greens
raspberry	blues
darksalmon	yellows
salmon	magentas
deepsalmon	cyans
warmpink	oranges
firebrick	tints
ruby	grays
chocolate	
brown	

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
CtSh Sele Orig Clip MoveZ
SnglClk +/- Cent Menu
DblClk Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

bg_rgb set to blue

```
SyntaxError: invalid syntax
PyMOL>set bg_rgb, blue
Setting: bg_rgb set to blue.
Setting: bg_rgb set to black.
PyMOL>select chain A and resid 20
Selector: selection "sele" defined with 12 atoms.
```

PyMOL> select chain A and resid 20

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild



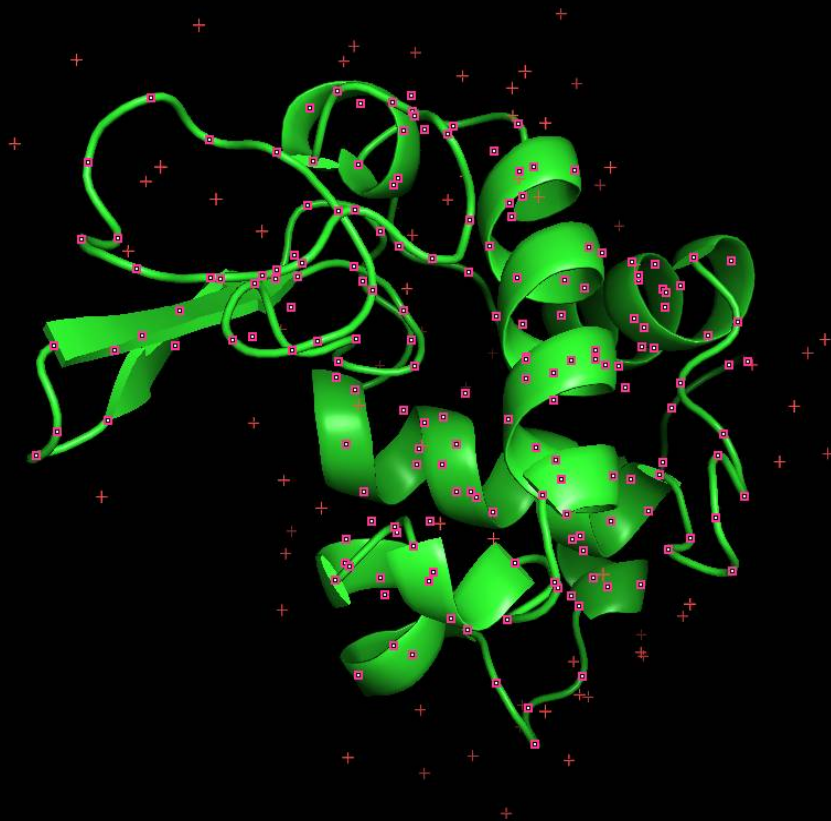
all	A	S	H	L	C
centered 1/1	A	S	H	L	C
(sele)	A	S	H	L	C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

CmdLoad: "" loaded as "centered".
Dragging whole object "centered".
Setting: transparency set to 0.50000.
Spectrum: range (15.29000 to 61.04000).
Executive: Colored 1082 atoms and 1 object.
Spectrum: range (17.45000 to 41.11000).
PyMOL>select resn ALA or name CA
Selector: selection "sele" defined with 177 atoms.

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL> select resn ALA or name CA



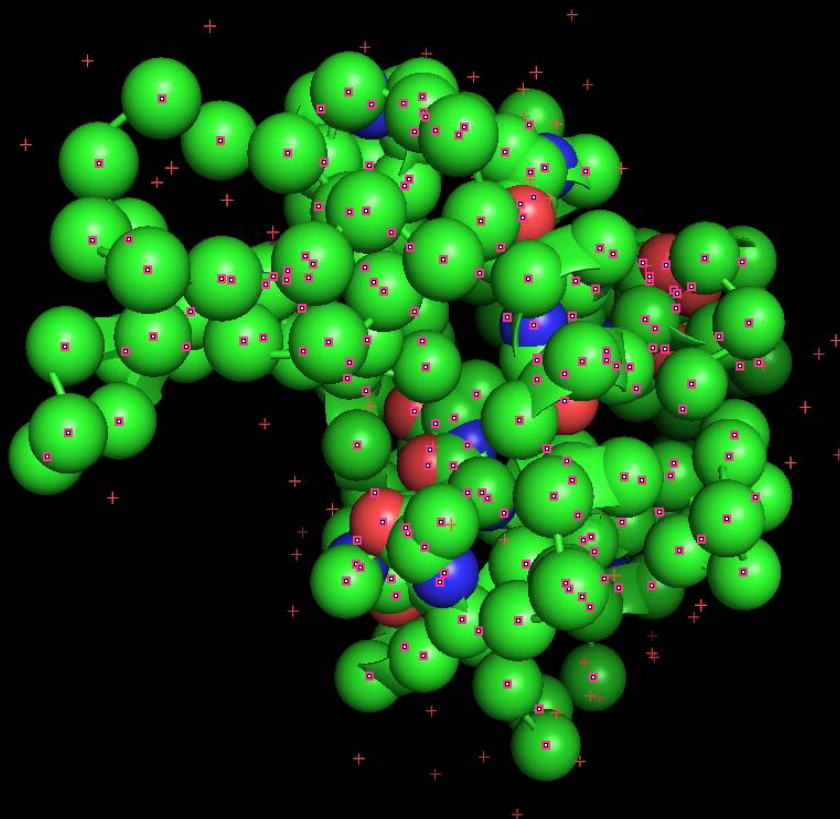
all	A	S	H	L	C
centered 1/1	A	S	H	L	C
(sele)	A	S	H	L	C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rota Move Mov2 Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MovS2
CtSh Sele Orig Clip MovZ
SnglClk +/- Cent Menu
DbClk Menu - PkAt
Selecting Residues
State 1/ 1

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

```
CmdLoad: "" loaded as "centered".  
Dragging whole object "centered".  
Setting: transparency set to 0.50000.  
Spectrum: range (15.29000 to 61.04000).  
Executive: Colored 1082 atoms and 1 object.  
Spectrum: range (17.45000 to 41.11000).  
PyMOL>select resn ALA or name CA  
Selector: selection "sele" defined with 177 atoms.
```

Reset	Zoom	Orient	Draw/Ray ▾			
Unpick	Deselect	Rock	Get View			
<	<	Stop	Play	>	>	MClear
Builder		Properties		Rebuild		

PyMOL> 

all	A	S	H	L	C
centered 1/1	A	S	H	L	C
(sele)	Show:				
as					
wire					
lines					
nonbonded					
licorice					
sticks					
nb_spheres					
ribbon					
cartoon					
label					
cell					
dots					
spheres					
mesh					
surface					
flag ignore					
organic					
main chain					
side chain					
disulfides					
valence					

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rota Move MovZ Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MvSZ
CtSh Sele Orig Clip MovZ
SnglClk +/- Cent Menu
DbClk Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

Spectrum: range (17.45000 to 41.11000).

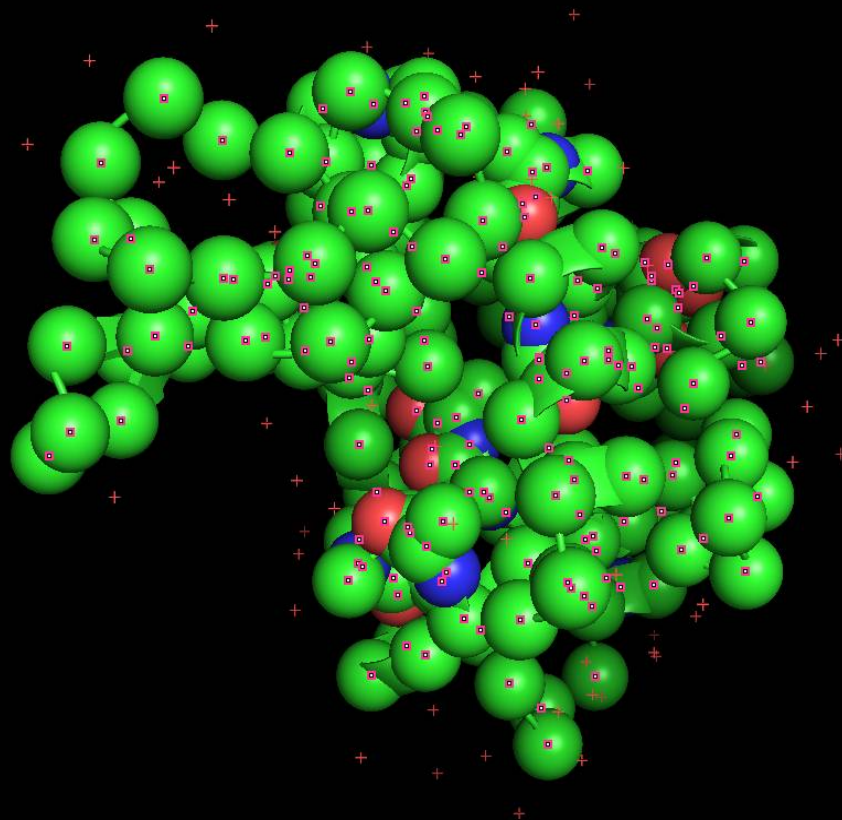
PyMOL>select resn ALA or name CA

Selector: selection "sele" defined with 177 atoms.

Incentive-Only-Error: "clean" is not available in Open-Source PyMOL

Please visit <http://pymol.org> if you are interested in the
full featured "Incentive PyMOL" version.

Reset	Zoom	Orient	Draw/Ray ▾
Unpick	Deselect	Rock	Get View
<	<	Stop	Play >
>		MClear	
Builder	Properties	Rebuild	

PyMOL> 

all	A S H L C
centered 1/1	A S H L C
(sele)	Action:
	delete selection
	rename selection
	zoom
	orient
	center
	origin
	drag coordinates
	clean
	modify
	preset
	find
	align
	remove atoms
	hydrogens
Copy To:	duplicate
new	copy to object
	extract object
	masking
	movement
	compute

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Shift Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

Spectrum: range (17.45000 to 41.11000).

PyMOL>select resn ALA or name CA

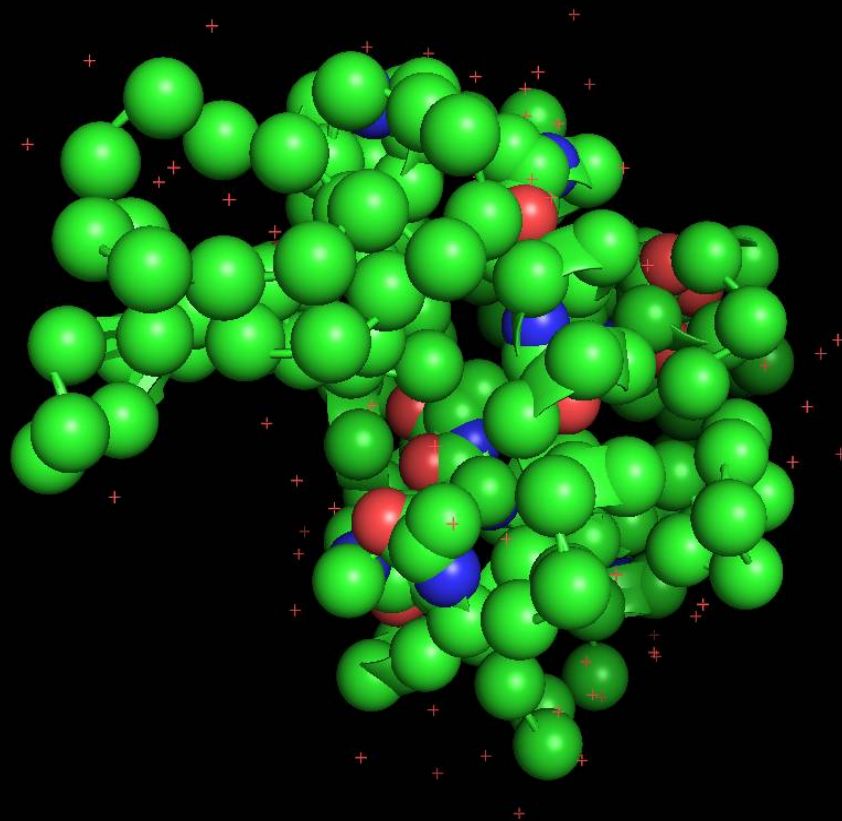
Selector: selection "sele" defined with 177 atoms.

Incentive-Only-Error: "clean" is not available in Open-Source PyMOL

Please visit <http://pymol.org> if you are interested in the full featured "Incentive PyMOL" version.

PyMOL>

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild



all	A	S	H	L	C
centered 1/1	A	S	H	L	C
(sele)	A	S	H	L	C
obj01 1/1	A	S	H	L	C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>select resn ALA or
Selector: selection "sele
Incentive-Only-Error: "c

Please visit <http://pymol.sourceforge.net>
full featured "Incentive-Only-Error: "c

Setting: bg_rgb set to w

PyMOL>

☐ Sequence
Sequence Mode
☒ Internal GUI
☒ Internal Prompt
Internal Feedback
Overlay
☐ Stereo
Stereo Mode
Zoom
Clip
Background
Color Space
Quality
Grid
☐ Orthoscopic View
☒ Show Valences
☒ Smooth Lines
☒ Depth Cue (Fogging)
☒ Two Sided Lighting
☒ Specular Reflections
☒ Animation
☐ Roving Detail
External GUI

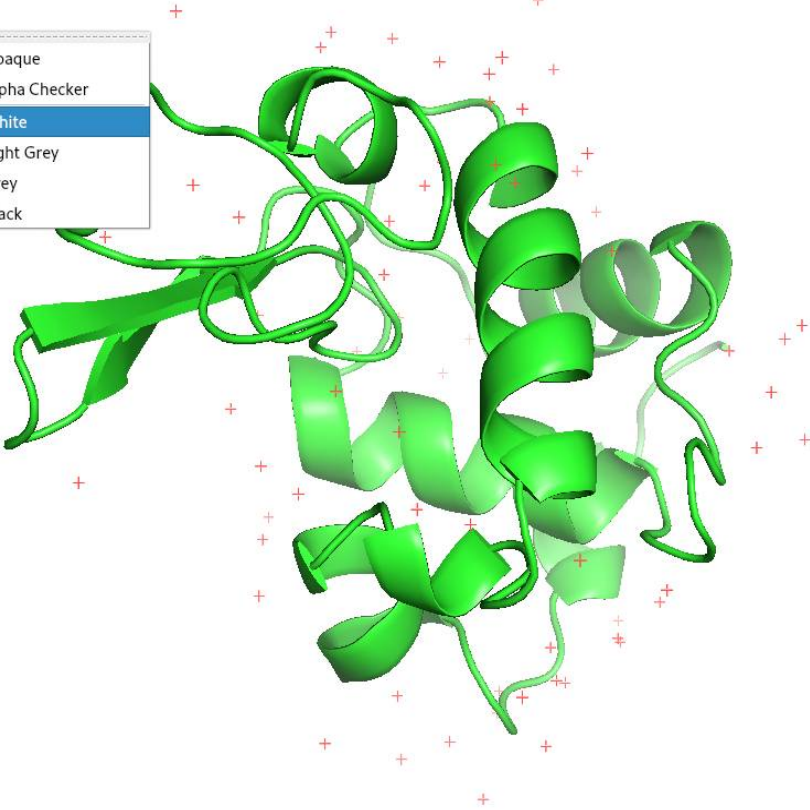
Source PyMOL
in the

☐ Opaque
☒ Alpha Checker
• White
Light Grey
Grey
Black

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

all A S H L C
centered 1/1 A S H L C

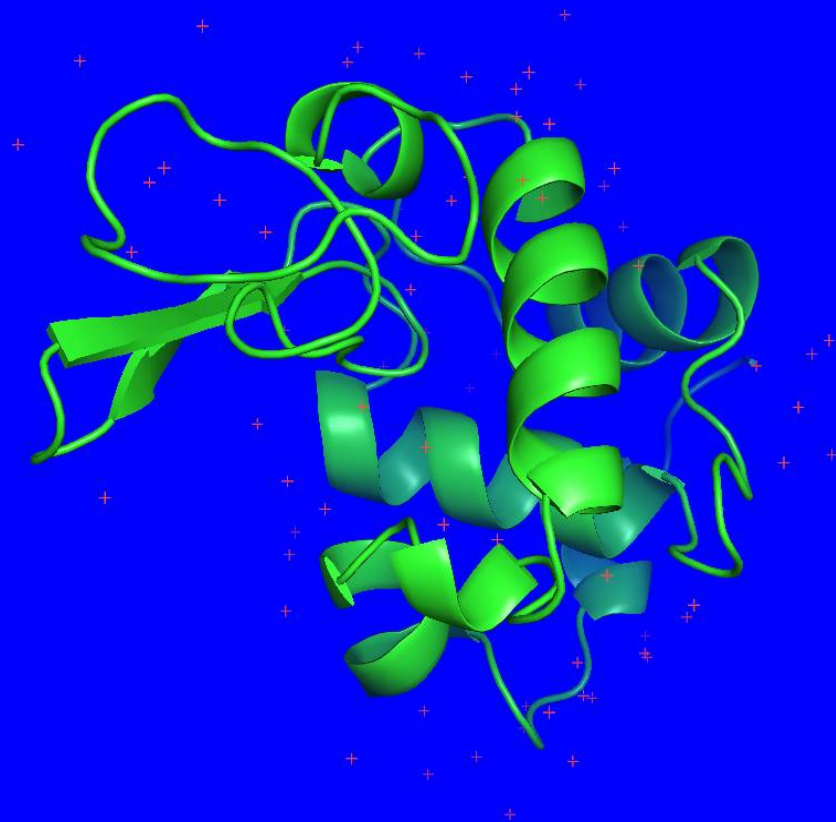
Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/1



Setting: bg_rgb set to black.
PyMOL>bg_rgb set to blue
File "toplevel", line 1
bg_rgb set to blue
SyntaxError: invalid syntax
PyMOL>set bg_rgb, blue
Setting: bg_rgb set to blue.

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > |> MClear
Builder Properties Rebuild

PyMOL> set bg_rgb, blue



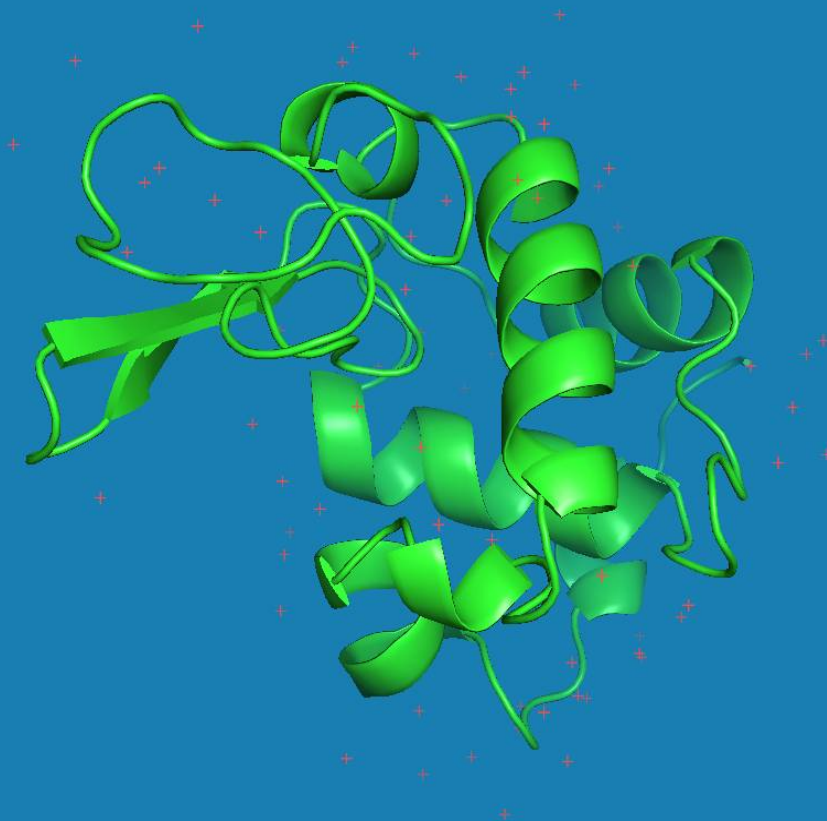
all A S H L C
centered 1/1 A S H L C

Mouse Mode 3-Button Viewing
Buttons L H R Wheel
e Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

GL RENDERER: Mesa Intel(R) Xe Graphics (TGL GT2)
GL_VERSION: 4.6 (Compatibility Profile) Mesa 20.3.5
Detected 8 CPU cores. Enabled multithreaded rendering.
TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".
PyMOL>set bg_rgb,[0.1,0.5,0.7]
Setting: bg_rgb set to 0x197fb2.

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL> set bg_rgb,[0.1,0.5,0.7]



all A S H L C
centered 1/1 A S H L C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rota Move Mov2 Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1 MvSZ
CtSh Sele Orig Clip MovZ
SnglClk +/- Cent Menu
DbIClk Menu - PkAt
Selecting Residues
State 1/ 1

```
SyntaxError: invalid syntax
PyMOL>set bg_rgb, blue
Setting: bg_rgb set to blue.
Setting: bg_rgb set to black.
PyMOL>select chain A and resid 20
Selector: selection "sele" defined with 12 atoms.
PyMOL>wizard measurement
```

PyMOL>

- Appearance
- Measurement
- Mutagenesis
- Pair Fitting
- Density
- Filter
- Sculpting
- Label
- Charge
- Demo

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

Please click on the first atom...



all A S H L C
centered 1/1 A S H L C

Measurement
Distances
Create New Object
Delete Last Object
Delete All Measurements
Done

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Shift Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Atoms
State 1/ 1

PyMOL>_

```
SyntaxError: invalid syntax
PyMOL>set bg_rgb, blue
Setting: bg_rgb set to blue.
Setting: bg_rgb set to black.
PyMOL>select chain A and resid 20
Selector: selection "sele" defined with 12 atoms.
PyMOL>wizard measurement
```

PyMOL>

Please click on the first atom...



Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

all A S H L C
centered 1/1 A S H L C

Measurement
Dis Measurement Mode
Cre Distances
Del Distances to Rings
Del Angles
Don Dihedrals
Mov Polar Neighbors
Bu Heavy Neighbors
& Neighbors
Polar Contacts
Lion Sele Orig Clip
SnglClk +/- Cent Menu
DblClk Menu - PkAt
Selecting Atoms
State 1/ 1

PyMOL>_

Setting: bg_rgb set to blue.
Setting: bg_rgb set to black.
PyMOL>select chain A and resid 20
Selector: selection "sele" defined with 12 atoms.
PyMOL>wizard measurement
You clicked /centered//A/HOH'130/0
Selector: selection "sele" defined with 1 atoms.
You clicked /centered//A/HOH'204/0

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>

Please click on the first atom...



all	A	S	H	L	C
sele	A	S	H	L	C
measure01	A	S	H	L	C

Measurement

Distances

Create New Object

Delete Last Object

Delete All Measurements

Done

Mouse Mode 3-Button Viewing

Buttons L M R Wheel

& Keys Rotate Move Move2 Slab

Shift +Box -Box Clip MoveS

Ctrl Move PkAt Pk1 MoveS2

Ctrl Sele Orig Clip MoveZ

SingleClick +/- Cent Menu

DoubleClick Menu - PkAt

Selecting Atoms

State 1/ 1

PyMOL>_

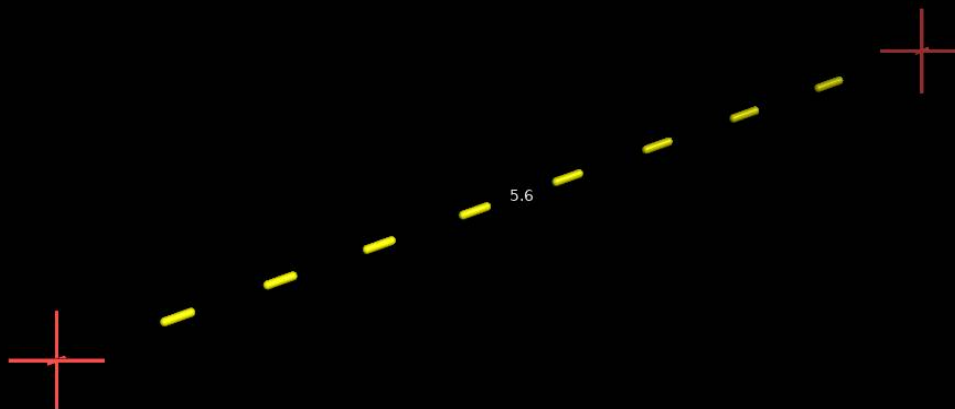
File Edit Build Movie Display Setting Scene Mouse Wizard Plugin Help

Setting: bg_rgb set to blue.
Setting: bg_rgb set to black.
PyMOL>select chain A and resid 20
Selector: selection "sele" defined with 12 atoms.
PyMOL>wizard measurement
You clicked /centered//A/HOH'130/0
Selector: selection "sele" defined with 1 atoms.
You clicked /centered//A/HOH'204/0

Reset Zoom Orient Draw/Ray ▾
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

PyMOL>

Please click on the first atom...



all	A	S	H	L	C
centered 1/1	A	S	H	L	C
measure01	Action: ▾				
	zoom				
	center				
	origin				
	drag				
	reset				
	rename				
	group				
	delete				

Measurement

Distances

Create New Object

Delete Last Object

Delete All Measurements

Done

Mouse Mode 3-Button Viewing

Buttons L M R Wheel

& Keys Rotate Move Move2 Slab

Shift +Box -Box Clip MoveS

Ctrl Move PkAt Pk1 MoveS2

CtSh Sele Orig Clip MoveZ

SingleClick +/- Cent Menu

DoubleClick Menu - PkAt

Selecting Atoms

State 1/ 1

PyMOL>_

|< < > >| S ▾ E

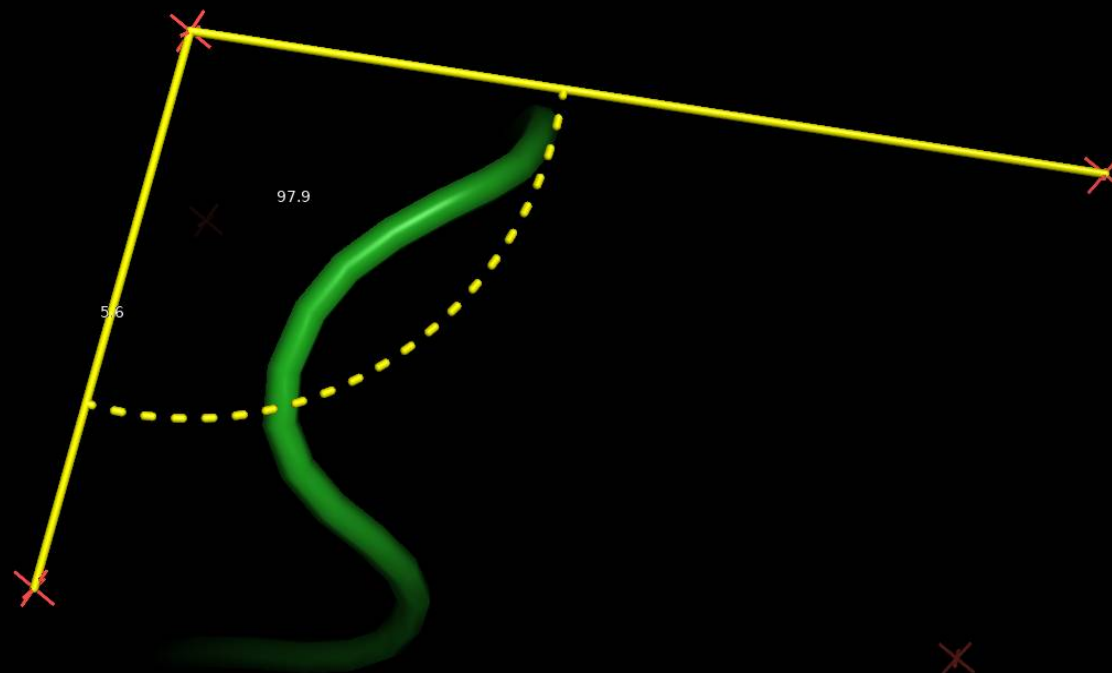
PyMOL>wizard measurement

You clicked /centered//A/HOH`130/0
Selector: selection "sele" defined with 1 atoms.
You clicked /centered//A/HOH`204/0
You clicked /centered//A/HOH`166/0
Selector: selection "sele" defined with 1 atoms.
You clicked /centered//A/HOH`204/0
You clicked /centered//A/HOH`130/0

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > |> MClear
Builder Properties Rebuild

PyMOL>

Please click on the first atom...



all	A	S	H	L	C
centered 1/1	A	S	H	L	C
measure01	A	S	H	L	C
measure02	A	S	H	L	C

Measurement

Angles

Create New Object

Delete Last Object

Delete All Measurements

Done

Mouse Mode 3-Button Viewing

Buttons L M R Wheel

& Keys Rotate Move Move2 Slab

Shift +Box -Box Clip MoveS

Ctrl Move PkAt Pk1 MvS2

Ctrl Sele Orig Clip MoveZ

SingleClick +/- Cent Menu

DoubleClick Menu - PkAt

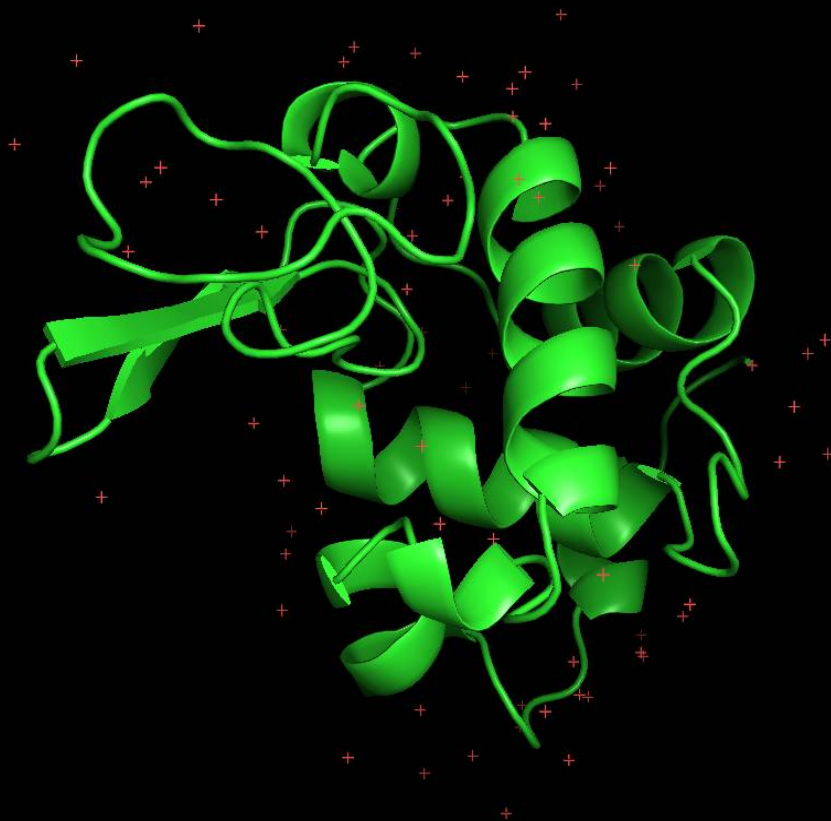
Selecting Atoms

State 1/ 1

PyMOL>_

OpenGL graphics engine:
GL_VENDOR: Intel
GL_RENDERER: Mesa Intel(R) Xe Graphics (TGL GT2)
GL_VERSION: 4.6 (Compatibility Profile) Mesa 20.3.5
Detected 8 CPU cores. Enabled multithreaded rendering.
TITLE LYSOZYME C
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "centered".

PyMOL>



Reset Zoom Orient Draw/Ray

Width 1700px 28.79cm

Height 802px 13.58cm

Reset Lock aspect ratio

Units cm at 150 DPI

transparent background ("Ray" only)

Draw (fast) - or - Ray (slow)

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

PyMOL>_

ObjectMolecule: Read crystal symmetry information.

CmdLoad: "" loaded as "centered".

PyMOL>ray 1700, 802, async=1

Ray: render time: 0.37 sec. = 9620.8 frames/hour (0.37 sec. accum.).

You clicked /centered//A/ASN 74/CA

Selector: selection "sele" defined with 8 atoms.

PyMOL>ray 1700, 802, async=1

Ray: render time: 0.37 sec. = 9814.0 frames/hour (0.74 sec. accum.).

PyMOL>

Reset Zoom Orient Draw/Ray ▾

Save Image to File

Copy Image to Clipboard

< Back



PyMOL>_

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
Ctrl Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1

OpenGL graphics engine:
GL_VENDOR: Intel
GL_RENDERER: Mesa Intel(R) Xe Graphics
GL_VERSION: 4.6 (Compatibility Profile) Mesa 23.0.1
Detected 8 CPU cores. Enabled multi-threading.
TITLE: LYSOZYME C
ObjectMolecule: Read crystal symmetry
CmdLoad: "" loaded as "centered".

PyMOL>

Edit All...
Keyboard Shortcuts...
Colors...

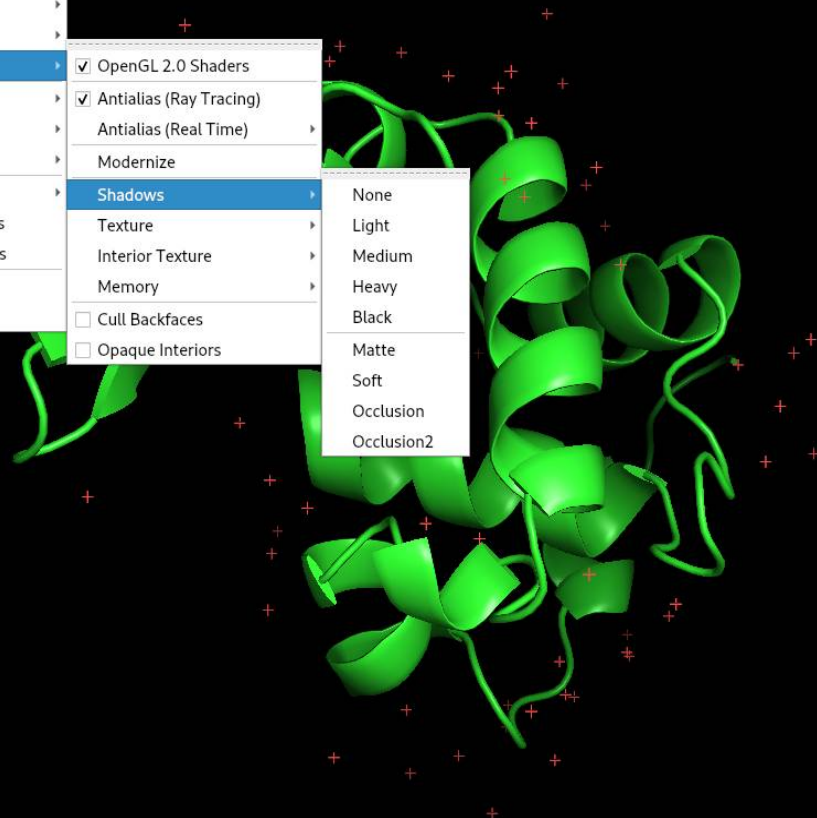
Label
Lines & Sticks
Cartoon
Ribbon
Surface
Volume
Transparency

Rendering
PDB File Loading
mmCIF File Loading
Map File Loading
Auto-Show ...
☒ Auto-Zoom New Objects
☐ Auto-Remove Hydrogens
☐ Show Text (Esc)
☐ Overlay Text

☒ OpenGL 2.0 Shaders
☒ Antialias (Ray Tracing)
☐ Antialias (Real Time)
☐ Modernize
Shadows
Texture
Interior Texture
Memory
☐ Cull Backfaces
☐ Opaque Interiors

None
Light
Medium
Heavy
Black
Matte
Soft
Occlusion
Occlusion2

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild



all A S H L C
centered 1/1 A S H L C

Mouse Mode 3-Button Viewing
Buttons L M R Wheel
& Keys Rotate Move Move2 Slab
Shift +Box -Box Clip MoveS
Ctrl Move PkAt Pk1 MoveS2
CtrlSh Sele Orig Clip MoveZ
SingleClick +/- Cent Menu
DoubleClick Menu - PkAt
Selecting Residues
State 1/ 1



Reset Zoom Orient Draw/Ray ▾

Width 1700px 28.79cm

Height 802px 13.58cm

Reset ☒ Lock aspect ratio

Units cm at 150 DPI

☒ transparent background ("Ray" only)

Draw (fast) - or - Ray (slow)

Use high-quality ray-tracing. Supports optimal multi-layer transparency, shadows, and alpha-channel background.

```

Mouse Mode 3-Button Viewing
Buttons      L      M      R      Wheel
& Keys      Rota  Move  MovZ  Slab
Shift +Box -Box Clip MovS
Ctrl Move PkAt Pk1  MvS2
CtSh Sele Orig Clip MovZ
SnglClk +/- Cent Menu
DbtClk Menu - PkAt
Selecting Residues
State 1/ 1

```

PyMOL>_

Applications: PyMOL

Terminal - dobrevp@no...

Builder

Chemical

Protein

Nucleic Acid

Ace

Ala

Arg

Asp

Cys

Gln

Glu

Gly

His

Ile

Leu

Lys

Met

Phe

Pro

Ser

Thr

Trp

Tyr

Val

NMe

NHH

Secondary Structure: Alpha Helix

Atoms: Fix H Add H Invert Delete Clear Charge: +1 0 -1 Residue: Remove

Bonds: Create Delete Cycle Arom Model: Clean Sculpt Fix Rest

☒ El-stat ☐ Bumps ☒ Undo Enabled Undo Redo

Pick locations to attach asp...

PyMOL

Reset

Zoom

Orient

Draw/Ray

Unpick

Deselect

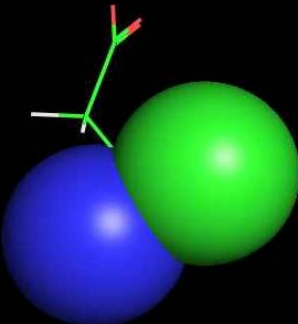
Rock

Get View

Builder

Properties

Rebuild



all

obj01 1/1

Creating Multiple Residues

Create As New Object

Buttons L M R Wheel

& Keys Rota Move MovZ Slab

Shft Rota MovD MuDZ MovS

Ctrl MovA +/- PKTB MvSZ

Ctrl MovAZ Orig Clip MovZ

SngrCik PkAt Cent Menu

DbicCik MovA DrGM PkTB

Picking Atoms (and Joints)

State 1/ 1

Builder

Chemical Protein Nucleic Acid

Ace Ala **Arg** Asn Asp Cys Gln Glu Gly His Ile Leu

Lys Met **Phe** Pro Ser Thr Trp Tyr Val NMe NHH

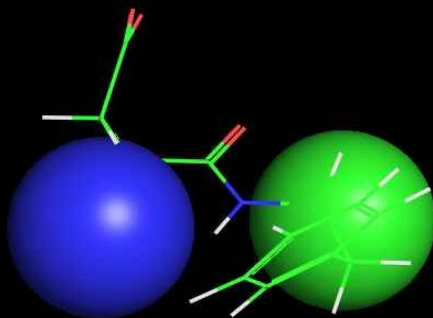
Secondary Structure: Alpha Helix

Atoms: Fix H Add H Invert Delete Clear Charge: +1 0 -1 Residue: Remove

Bonds: Create Delete Cycle Arom Model: Clean Sculpt Fix Rest

☒ El-stat ☐ Bumps ☒ Undo Enabled Undo Redo

Pick locations to attach phe...



Reset Zoom Orient Draw/Ray

Unpick Deselect Rock Get View

|< < Stop Play > >| MClear

Builder Properties Rebuild

all A S H L D

obj01 1/1 A S H L D

Attaching Multiple Residues

Create As New Object

Done

Mouse Mode 3-Button Editing

Buttons L M R Wheel

& Keys Rota Move MovZ Slab

Shft Rota MovD MuDZ MovS

Ctrl MovA +/- PKTB MvSZ

Ctsh MovAZ Orig Clip MovZ

SngrCik PkAt Cent Menu

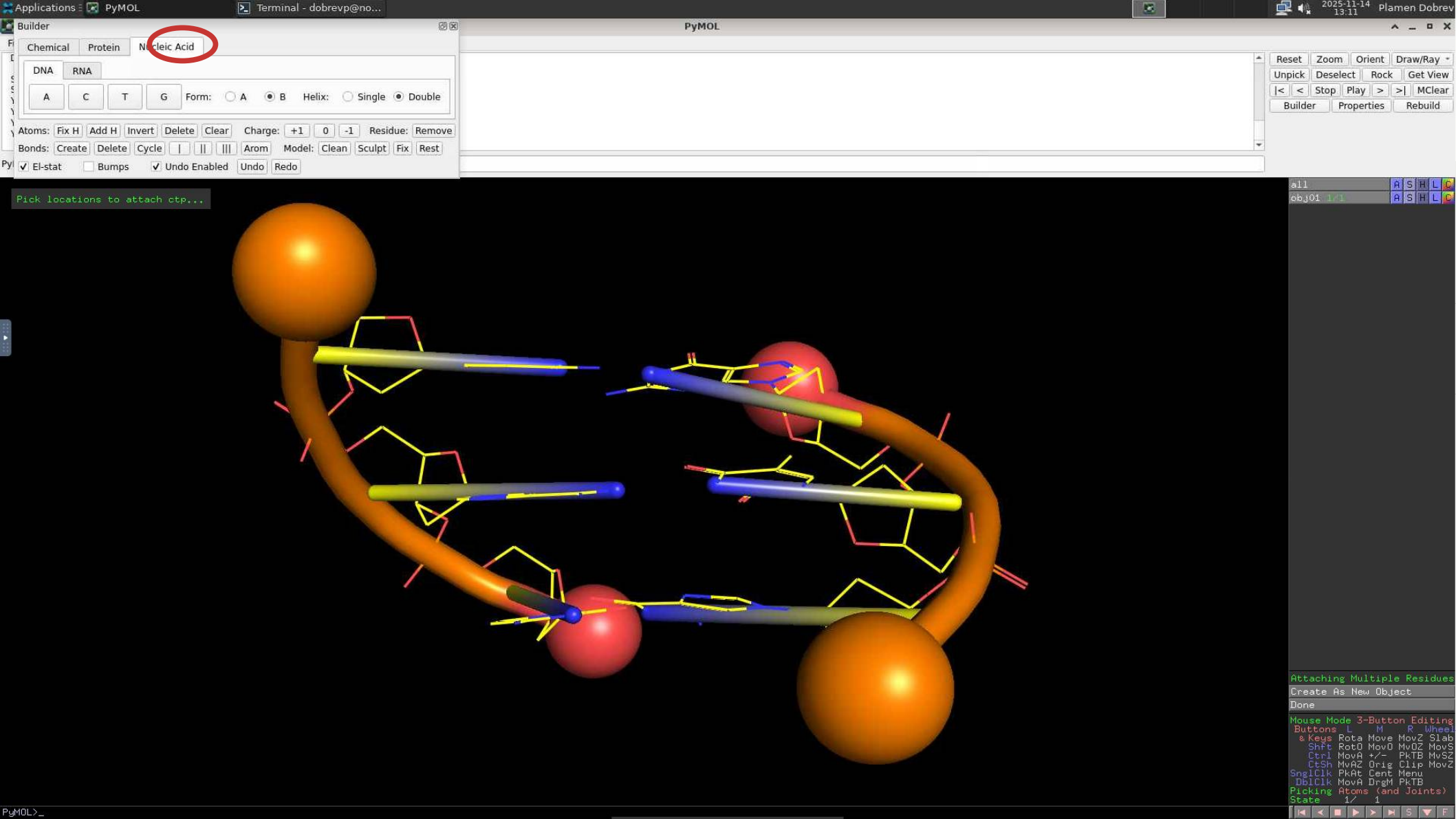
DbicCik MovA DrGM PKTB

Picking Atoms (and Joints)

State 1/ 1

You clicked /obj01///ASP*2/C -> (pk1)

PyMOL>_



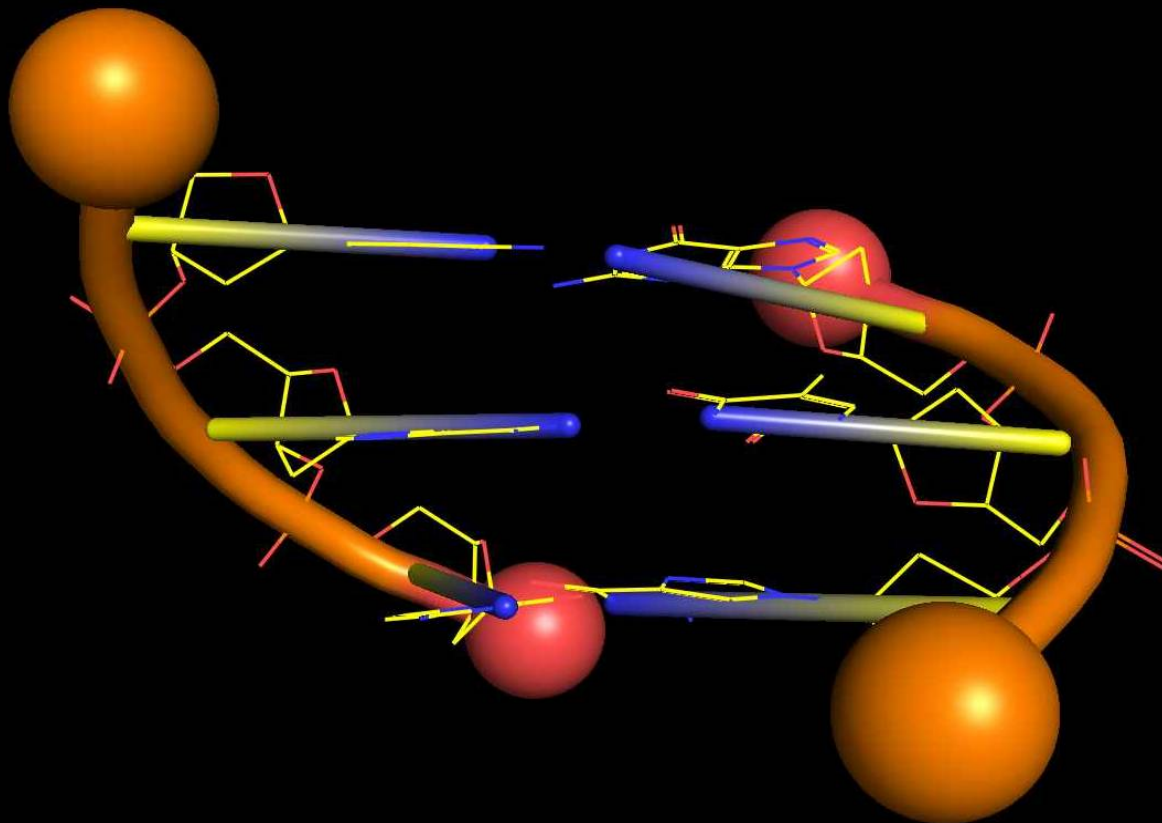
- New PyMOL Window
- Open...
- Open Recent...
- Get PDB...
- Save Session
- Save Session As...
- Export Molecule...
- Export Map...
- Export Alignment...
- Export Image As
- Export Movie As
- Log File
- Run Script...
- Working Directory
- Edit pymolrc
- Reinitialize
- Quit

abled multithreaded rendering.

to off.
ndo is not supported in Open-Source version of PyMOL
/N -> (pk1)
/C -> (pk1)
-1/P -> (pk1)
1/P -> (pk1)

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
|< < Stop Play > >| MClear
Builder Properties Rebuild

ch ctp...



all
obj01 1/1

Attaching Multiple Residues
Create As New Object
Done

Mouse Mode 3-Button Editing
Buttons L M R Wheel
& Keys Rota Move MovZ Slab
Shft Rot0 Mov0 MovZ MovS
Ctrl MovA +/- PKTB MovS2
CtSh MovAZ Orig Clip MovZ
SnglClk PkAt Cent Menu
DbClk MovA DrgM PKTB
Picking Atoms (and Joints)
State 1/ 1

Detected 80 CPU cores. Enabled multithreaded rendering.

Setting: suspend_undo set to off.
Setting-Warning: suspend_undo is not supported in Open-Source version of PyMOL
You clicked /obj01///ASP`2/N -> (pk1)
You clicked /obj01///ASP`2/C -> (pk1)
You clicked /obj01/B/B/DT`-1/P -> (pk1)
You clicked /obj01/A/A/DA`1/P -> (pk1)

PyMOL>

Pick locations to attach ctp...

Reset Zoom Orient Draw/Ray
Unpick Deselect Rock Get View
< < Stop Play > > MClear
Builder Properties Rebuild



all	A	S	H	L
obj01 1:1	A	S	H	L

Attaching Multiple Residues

Create As New Object

Done

Mouse Mode 3-Button Editing

Buttons L M R Wheel

& Keys Rota Move MovZ Slab

Shft Rota Mov0 MovZ MovS

Ctrl MovA +/- PkTB MovSZ

Ctrl MovAZ Orig Clip MovZ

SingleClick PkAt Cent Menu

DoubleClick MovA Drag PkTB

Picking Atoms (and Joints)

State 1/ 1

PyMOL>